

Purchase Order Management System Using SAP S/4HANA(SAP MM)

Project submitted to the

SRM University – AP, Andhra Pradesh

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1. Introduction

SAP S/4HANA is an Enterprise Resource Planning (ERP) solution developed by SAP to manage various business operations such as procurement, production, sales, finance, and inventory management. It enables organizations to integrate their business processes into a single system, improving efficiency and decision-making.

The Materials Management (MM) module in SAP S/4HANA is responsible for procurement, inventory management, and material planning. It helps organizations manage purchasing activities, maintain vendor relationships, monitor inventory levels, and ensure the timely availability of materials.

This project demonstrates the complete procurement process using SAP MM, starting from creating master data to completing invoice verification

2. Project Objective

The objective of this project is to implement the complete Purchase-to-Pay (P2P) process using SAP S/4HANA Materials Management (MM). The project aims to understand the procurement lifecycle, create purchasing documents, manage inventory transactions, and verify supplier invoices through standard SAP transactions.

3. Problem Statement

In many organizations, procurement activities are performed manually, leading to delays, data inconsistencies, duplicate records, and inefficient inventory management. A centralized ERP system is required to automate purchasing activities, maintain accurate inventory records, and streamline communication between departments

4. Proposed Solution

The proposed solution is a Purchase Order Management System implemented using SAP S/4HANA MM. The system automates procurement activities by maintaining vendor and material master data, creating purchase requisitions, generating purchase orders, recording goods receipts, and verifying supplier invoices. This ensures efficient purchasing operations and accurate inventory management.

6. SAP Module Used

SAP Materials Management (MM)

SAP MM is one of the core modules in SAP ERP that manages procurement and inventory functions. It supports purchasing activities, material management, inventory control, and vendor management. The module integrates with Finance (FI), Production Planning (PP), and Sales & Distribution (SD), enabling smooth business operations.

7. Business Scenario

A company regularly purchases laptops and office equipment from authorized vendors. Whenever a department requires materials, it raises a Purchase Requisition. The purchasing department reviews the request and creates a Purchase Order for the selected vendor. After the materials are delivered, the warehouse records the Goods Receipt, updating the inventory. Finally, the supplier invoice is verified before payment is processed. This project simulates the complete procurement process using SAP MM.

8. Procurement Process Flow

The procurement cycle begins with creating the Vendor Master and Material Master. A Purchase Requisition is generated based on business requirements. The Purchase Requisition is converted into a Purchase Order and sent to the vendor. Once the materials are received, a Goods Receipt is posted to update inventory. Finally, Invoice Verification ensures that the supplier invoice matches the purchase order and goods receipt before payment approval.

9. Transaction Codes Used

Transaction Code	Description
BP	Create Business Partner (Vendor)
MM01	Create Material Master
ME51N	Create Purchase Requisition
ME21N	Create Purchase Order
MIGO	Post Goods Receipt
MIRO	Invoice Verification
ME23N	Display Purchase Order

10. Vendor Master Creation (BP)

The Vendor Master stores all the essential information about suppliers, including their name, address, contact details, payment terms, and purchasing information. Creating an accurate vendor master ensures that procurement transactions are processed correctly.

11. Material Master Creation (MM01)

The Material Master contains information about materials purchased and stored by the organization. It includes details such as material number, description, unit of measure, material type, valuation data, and storage information. This data is used throughout the procurement and inventory processes.

12. Purchase Requisition (ME51N)

A Purchase Requisition is an internal request generated by a department to procure specific materials. It specifies the required material, quantity, and delivery date. The Purchase Requisition serves as the starting point of the procurement process.

13. Purchase Order Creation (ME21N)

A Purchase Order is an official document issued to the vendor requesting the supply of materials. It includes vendor details, material information, quantity, pricing, delivery schedule, and payment terms. Once approved, the Purchase Order is sent to the supplier.

14. Goods Receipt (MIGO)

The Goods Receipt transaction confirms that the ordered materials have been received from the vendor. Posting the Goods Receipt updates inventory quantities and records the receipt of materials in the SAP system.

15. Invoice Verification (MIRO)

Invoice Verification compares the supplier invoice with the Purchase Order and Goods Receipt. SAP performs a three-way match to ensure that the invoice quantity and amount are correct before payment is approved.

16. Purchase Order Display (ME23N)

The Purchase Order Display transaction allows users to view complete Purchase Order details, including vendor information, ordered materials, pricing, delivery status, and

document history. It helps users verify procurement documents and track purchasing activities.

ScreenShots:

<https://drive.google.com/file/d/10vQLOIteoKgwD4HUSrxAmemo-Yu6TZGh/view?usp=sharing>

17. Results

The Purchase Order Management System was successfully implemented using SAP S/4HANA MM. The complete procurement cycle was executed, including vendor creation, material creation, purchase requisition, purchase order generation, goods receipt, and invoice verification. The project demonstrated how SAP automates procurement activities while maintaining accurate inventory and purchasing records.

18. Advantages

- Automates the procurement process.
- Reduces manual documentation.
- Improves inventory accuracy.
- Enhances vendor management.
- Ensures transparency in purchasing.
- Supports faster decision-making.
- Reduces procurement errors.
- Maintains centralized business data.

19. Limitations

- Requires trained SAP users.
- Initial implementation cost is high.

- Customization requires technical expertise.
- Dependent on accurate master data.
- Licensing costs can be significant for organizations.

20. Future Enhancements

- Implement automated purchase approval workflows.
- Integrate with SAP FICO for automatic payment processing.
- Develop vendor performance evaluation reports.
- Create procurement dashboards using SAP Analytics.
- Enable mobile approval of purchase orders.
- Integrate AI-based demand forecasting for procurement planning.

21. Conclusion

The Purchase Order Management System successfully demonstrates the end-to-end procurement process using SAP S/4HANA Materials Management. The project provides practical knowledge of procurement activities, inventory management, and invoice verification. It highlights the benefits of ERP systems in improving operational efficiency, reducing manual effort, and ensuring accurate business transactions. This project also strengthened my understanding of SAP MM concepts and their application in real-world business scenarios.